SymPy Bug Card

Bug 52: `function_range` Says `sqrt(tan(x)**2)` Is Constant

Task. Compute the real range of the expression on the interval $[-1, 1]$.

$$\text{function\_range}\Big(\sqrt{\tan(x)^2}, x, [-1, 1]\Big)$$

SymPy's answer.

$$\text{function\_range}\Big(\sqrt{\tan(x)^2}, x, [-1, 1]\Big) = \{\tan(1)\}.$$

Correct answer.

$$[0, \tan(1)].$$

Explanation. On $[-1, 1]$, $\sqrt{\tan(x)^2} = |\tan(x)|$. The function is continuous there, has value 0 at $x = 0$, and has maximum $\tan(1)$ at both endpoints, so the singleton answer omits attained values.

Likely root cause. The diagnosis points to `sympy/calculus/util.py`: `function_range` samples endpoints and derivative zeros, but misses points where the derivative is undefined while the original function remains continuous. This is a new instance of a known failure mode.

Artifact (GitHub).
`bug-report/bug-052-function-range-says-sqrt-tan-x-2-is-constant/`